# MARY SEMERDJIAN

Los Angeles, CA | Email | Portfolio


## EDUCATION

**California State University, Los Angeles** **May 2024**
*Master of Science in Computer Science, GPA: 3.90/4.0* *Los Angeles, CA*
Thesis: Machine Learning tools for Real-World DDoS detection.
Coursework: Computer Graphics, Data Science, Advanced Software Engineering - Architecture, Advanced Topics in Operating Systems – Artificial Intelligence – Functional Programming– Web Programming, Computer and Network Security, Computer Networks Configuration and Management

**California State University, Los Angeles** **May 2022**
*Bachelor of Science in Computer Science* *Los Angeles, CA*
Coursework: Programming Language Paradigms, Analysis of Algorithms, Human-Centered Computing, Cryptography, Machine Learning - Deep Learning, Concurrent - Distributed Programming, Principles of Database Systems, Software Design Laboratory I-II, Computer Organization, Linear Optimization, and Theory of Probability.

**Los Angeles City College** **June 2019**
*Associate of Arts in Computer Science - Computer Information Systems, Liberal Arts - Natural Sciences and Mathematics, Arts and Humanities, Associate of Science in Physics for Transfer* *Los Angeles, CA*
Coursework: Intro to Data Structures, Programing Logic and Design, Programming in Java and C++, Calculus with Analytic Geometry I-II-III, Linear Algebra, Discrete Mathematics, Physics for Engineers and Scientists I-II-III, Beginning Photography, Intro to Photoshop, Biology, General Chemistry, and General Psychology I.

## PROFESSIONAL EXPERIENCE

**Los Angeles City College** **May 2025 – June 2025 & Sep 2025 – Dec 2025**
*Data Science Teaching Assistant* *Los Angeles, CA*

- Led data science lectures, guided 30+ students in Python and R projects applying Machine Learning models to real-world datasets.

**Los Angeles City College** **June - Aug 2025**
*Calculus Teaching Assistant* *Los Angeles, CA*

- Taught Calculus I, provided technical instruction and tutoring to 40+ students.

**California State University, Los Angeles** **Aug – Dec 2023**
*Graduate Assistant – Computer Network Security* *Los Angeles, CA*

- Delivered lab sessions on threat analysis, firewall configuration, and vulnerability scanning to 60+ graduate students.

## PROFESSIONAL PROJECTS & RESEARCH

**Applying Machine Learning Tools for Real-World DDoS Detection (Master's Thesis)** **Aug 2023 – May 2024**
Technologies: Big Data, Python, Machine Learning, High Performance Computing, Neural Networks, CatBoost, Scikit-Learn, Exploratory Data Analysis

- Conducted supervised research under a faculty advisor on Machine Learning approaches for detecting and mitigating real-world Distributed Denial of Service (DDoS) attacks.
- Applied anomaly detection, classification, and ensemble methods to improve detection accuracy.
- Processed large-scale network traffic datasets using High Performance Computing (HPC) infrastructure.

**Applying Advanced Regression Models and Machine Learning for Differentiated Thyroid Cancer Patient with Recurrences** **Feb – Jun 2025**
Technologies: RStudio, R, Google Colab, Scikit-learn, Pandas, Exploratory Data Analysis

- Applied advanced regression and Machine Learning classification to the Differentiated Thyroid Cancer Recurrence dataset from the UCI.
- Compared Generalized Linear Models (Binary Logistic, Probit, Complementary Log-Log) and Machine Learning Model (Random Forest) performance, reported findings in academic format.

**Data Science for Business Analytics: Intro to Google Analytics, Adobe Analytics, Microsoft Power BI** **Jan 2023 – May 2023**
Technologies: Adobe Analytics, Google Analytics, Microsoft Power BI, Hyperlinks

- Use of data, statistical algorithms, and technology to gain insights and drive better decision-making in both large and small companies.

**Honeypot Server** **Aug – Dec 2022**
Technologies: Ubuntu, Kali Linux, Windows OSX, Wireshark, KFSensor, Command Prompt, Root in Kali, Nmap

- Configured and deployed a Honeypot server to capture, analyze, and monitor malicious network traffic patterns in Kali Linux, Ubuntu, and Windows.
- Generated threat intelligence reports based on attack traffic behavior logs.

**Pelvic Image Analysis and Geometry Reconstruction Using Artificial Intelligence (Senior Capstone)** **Aug 2021 – May 2022**
Technologies: MRI image, 3D Slicer, NVIDIA AIAA, Docker, AWS, MONAI, Pytorch

- Led a front-end team of 5 undergraduate students, instructing them in 3D modeling fundamentals and AI-based medical image visualization.
- Integrated NVIDIA AIAA with 3D Slicer for automated segmentation of pelvic imaging data.

## TECHNICAL SKILLS

**Programming Languages:** C++, C, Java, JavaScript, Python, Haskell, Kotlin, BASIC, Swift, SAS, R, SQL
**IT / Cloud / DevOps:** Git, Docker, PowerShell, Bash, AWS, Windows, macOS, Photoshop, OpenGL, Linux (Kali/Ubuntu)
**Databases/Frameworks/Libraries:** MariaDB, MongoDB, phpMyAdmin, MySQL, DBeaver, Node.js, Express.js, Django, Bootstrap, NPM
**Data Science / Machine Learning:** Jupyter Notebook, TensorFlow, Pandas, Numpy, Keras, Matplotlib, Seaborn, PyTorch, XGBoost, Deep Learning, Scikit-Learn
**Analysis / Visualization:** Microsoft Excel, Google Analytics, Microsoft Power BI, Tableau, Adobe Analytics
**Cybersecurity:** Cisco, Nmap, Metasploit, Snort, Wireshark, Nessus, Firewall Configuration, Vulnerability Scan, Threat Detection, Penetration Testing, Kali Linux
**Soft Skills:** Team Leadership, Technical Writing, Technical Documentation, Analytical Thinking, Testing, Agile Methodology, Design Modification, Creativity, Critical Thinking, Algorithmic Solutioning, Analytical Problem-Solving, Design Thinking

## CERTIFICATIONS

**Cloud Summit Solutions Track -** SANS Institute (August 2026)
**Basic Medical Terminology / Teamwork, Collaboration, and Conflict Resolution -** LA Community College District (February 2026)
**Professionalism Integrity and Ethics / Customer Service Compassion and Empathy -** LA Community College District (February 2026)
**Cybersecurity Job Simulation** - Mastercard (August 2025) & **Cyber Job Simulation** – Deloitte (August 2025)
**Data Science Job Simulation** - British Airways (August 2025)
**Robotics and Control Job Simulation** - Johnson & Johnson MedTech (July 2025)
**Leveraging LLMs: Building & Securing RAG, Contextual RAG, and Agentic RAG** - SANS Institute (April 2025)

## VOLUNTEERING

**Jewel City Hacks 4.0** | Glendale Community College, CA — May 2026
**Grad Fair** | California State University, Los Angeles — March 2026
**American Heart Association's Greater Los Angeles Heart and Stroke Walk** | Pasadena, CA — September 2025
**Food Forward** | Santa Monica, CA — August 2025
**Holly J Mitchell Juneteenth Celebration and Resource Fair** | Roy Campanella Park, Compton, CA — June 2025